Numerical Solutions to PDEs

ECM3446/ECMM461: High Performance Computing

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Module Overview

Week 1: Introduction to HPC	Week 5: Floating point arithmetic	Week 8: Task parallelism and manger worker parallelism
Week 2: Introduction to OpenMP	Week 6: Memory and cache	Week 9: Linear algebra and matrices
Week 3: Numerical solutions to PDEs	Week 7: Factors affecting parallel performance	Week 10: Accelerators and GPUs
Week 4: Introduction to MPI		Week 11: Summary

Syllabus Plan

- Motivation of and introduction to high-performance computing
- Parallel computer architectures: shared-memory and distributed-memory architectures, multi-core processors, Graphics Processing Units (GPUs)
- Parallel programming methods: OpenMP, Message Passing Interface (MPI)
- Floating point arithmetic: floating point model, range, accuracy, exceptions
- Parallel processing algorithm and programming design: domain decomposition, halo exchange, manager-worker, task-based parallelism
- Parallel performance: speed-up, efficiency, parallel overheads and scaling
- Interconnection networks in high performance computers: topologies, latency and bandwidth

Numerical solutions to Partial Differential Equations

- Many HPC applications involve calculating numerical solutions to PDEs
- We saw examples in unit 1.1 (fluid dynamics calculations in weather/climate, astrophysics, engineering)
- We will focus on PDEs which vary in time and space
- We will look at three example equations
- We will also look at stability and accuracy of the solutions

Lecture Units

- Unit 3.1: Introduction to PDEs
- Unit 3.2: The exponential decay equation
- Unit 3.3: The advection equation
- Unit 3.4: The diffusion equation
- Unit 3.5: Accuracy
- Unit 3.6: Stability

Workshop 3

- Workshop 3 will be on 03rd/04th February.

- Workshop 3 looks at numerical solutions to the advection equation

- Before the workshop you will need to get familiar with units 3.1-3.3 and 3.5-3.6